A STUDY TO ASSESS CONTINUOUS GLUCOSE SENSOR PROFILES IN HEALTHY NON-DIABETIC PARTICIPANTS AGED <7 YEARS

PROTOCOL

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CHAPTER 1: INTRODUCTION

1.1 Background and Rationale

Studies aimed at improving glycemic control or preventing hypoglycemia often use continuous glucose monitoring (CGM)-measured percentages of glucose sensor values within, above or below glucose targets and glucose variability as outcome measures. Knowledge of the expected frequencies of each metric in a nondiabetic population is important for comparative purposes. The JDRF CGM RCT Study group previously assessed sensor profiles in a population of 30 non-diabetic participants in each of 4 age groups (i.e., 8 to <15, 15 to <25 and 25-45 and >45 years of age) using older generation sensors including the Dexcom STS Continuous Glucose Monitoring System™ (Dexcom, Inc., San Diego, CA), the Paradigm®/Guardian® REAL-Time Continuous Glucose Monitoring Systems (Medtronic Minimed, Northridge, CA) and the FreeStyle Navigator™ Continuous Glucose Monitoring System (Abbott Diabetes Care, Inc.)(1). Sensory accuracy has improved substantially since that time. More recently, data was collected on healthy individuals age 7 and up, using the Dexcom G6 (2). However, no data currently exists on young individuals <7 years.

It is vital to have data on this population as well to establish a reference range of sensor values in this age group. For example, determination of the frequency of sensor-derived, interstitial glucose concentrations that are <70 mg/dl in healthy participants using current generation sensors will be particularly important in assessing the ability of closed loop systems to prevent hypoglycemia in patients with T1D.

The objective of this protocol is to establish reference sensor glucose ranges in young children.

1.2 Protocol Synopsis

1.2.1 Study Objective

The primary objective of this study is to establish reference sensor glucose ranges in healthy, non-diabetic young children using a current generation CGM sensor.

1.2.2 Study Design

This will be a multi-center, observational study and will consist of up to 10 days of CGM sensor wear.

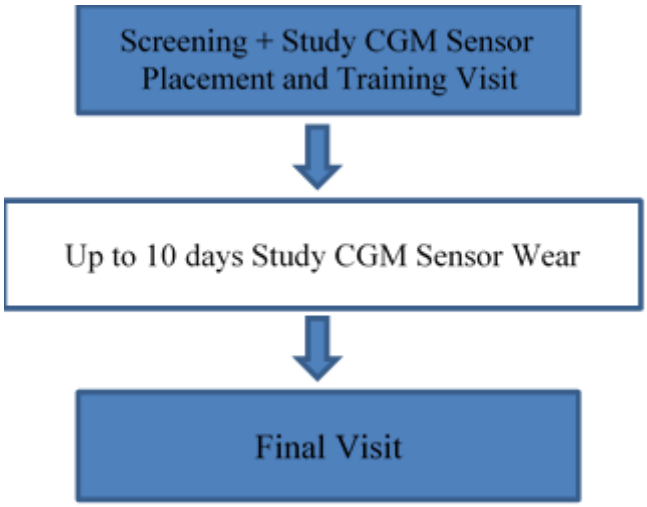
1.2.3 Sample Size

Approximately 50 individuals

1.2.4 Major Eligibility Criteria

1. Age 1 - <7 years
2. Body mass index (BMI) between the 10th to 90th percentile for age and sex
3. No history of diabetes - participants with point-of care (POC) HbA1c ≥5.7% will be excluded

1.3 Study Flow Chart



** If there are less than 72 hours of sensor data available and the participant is willing to wear the sensor again, an additional sensor will be inserted. If there are less than 72 hours of data and it is not repeated, the participant may be replaced. A participant may also be asked to return for repeat wear and repeat POC HbA1c if unexpected sensor values are returned.*

1.4 General Considerations

The study is being conducted in compliance with the policies described in the study policies document, with the ethical principles that have their origin in the Declaration of Helsinki, with the protocol described herein, and with the standards of Good Clinical Practice.

The Dexcom G6 CGM used in this study is FDA approved. Study participants will wear a blinded CGM for 10 days. Study participants do not have diabetes and the glucose values from the blinded CGM will not be used in any medical management. Therefore, the risk level is considered to be research not involving greater than minimal risk.

CHAPTER 2: ELIGIBILITY

2.1 Study Population

Approximately 50 healthy, nondiabetic individuals aged 1 to <7 years. Participants not completing the study may be replaced. Participants with a laboratory HbA1c of ≥5.7% or positive diabetes antibodies also may be replaced.

2.2 Eligibility

To be eligible for the study, a participant must meet the following inclusion criteria and none of the exclusion criteria:

2.2.1 Inclusion Criteria

1. Age 1 - <7 years
2. Body mass index (BMI) between the 10th to 90th percentile for age and sex

2.2.2 Exclusion Criteria

1. History of diabetes
2. Point of care (POC) HbA1c ≥5.7%
3. The presence of a significant medical disorder that in the judgment of the investigator will affect the wearing of the sensors, glucose metabolism, or the completion of any aspect of the protocol
4. The use of any steroid or other medication that in the judgment of the investigator will affect the wearing of the sensors, glucose metabolism, or the completion of any aspect of the protocol
5. Participation in another pharmaceutical drug or device study at the time of enrollment or during the study

2.3 Informed Consent

The study will be presented to and discussed with potential participants and their parent/legal guardians. The parent/legal guardians will be provided with the Informed Consent Form to read and will be given the opportunity to ask questions before deciding to participate and signing the Consent.

As part of the informed consent process, the parent or legal guardian of each participant will be asked to sign an authorization for release of personal information. The investigator, or his or her designee, will review what study specific information will be collected and to whom that information will be disclosed. After speaking with the parent, questions will be answered about the details regarding authorization.

CHAPTER 3: STUDY PROCEDURES

3.1 Initial Visit

3.1.1 Initial Study Procedures

After informed consent is signed, a history will be elicited (including family history of diabetes), a limited physical examination will be performed, and a POC HbA1c measurement will be obtained to verify eligibility. The following laboratory samples will be obtained. Participants can proceed in the study prior to the results being available; however, the results will be used to determine inclusion in the analyses.

- HbA1c
- Anti-GAD, anti-IA2, anti-insulin antibodies, and zinc-transporter-8 autoantibodies

3.1.2 Sensor Placement and Training

The device will be blinded so that the participant and the parent are not able to see the sensor readings. The participant and parent will be instructed on use of the CGM and the sensor will be inserted and initiated by the study personnel. The sensor will be worn for up to 10 days.

3.2 CGM Sensor Wear Period

3.2.1 CGM

Participants will be instructed to use the CGM for up to 10 days.

3.2.2 Completion of Home Log

The participant (parent) will be instructed to complete a log each day indicating sleep and wake times.

3.3 Final Visit

The participant (and parent) will return approximately 10 days after the blinded CGM sensor was placed.

The sensor will be removed and the CGM data will be downloaded.

*This visit may be completed remotely at investigator discretion.

The CGM data will be reviewed to assess whether the participant has used the CGM for at least 72 hours (equivalent to 3 days).

- Participants not meeting the CGM usage requirement or with unexpected sensor values may be given a second opportunity to wear the blinded CGM sensor at investigator discretion.
- If there are less than 72 hours of data and it is not repeated, the participant may be replaced.

CHAPTER 4: ADVERSE EVENTS AND RISKS

4.1 Adverse Event Reporting and Monitoring

4.1.1 Definition

For this study, only the following adverse events (AE) will be collected and reportable:

1. Any adverse event that meets criteria for a serious adverse event
2. Any device-related AE listed below:
   - Excessive pain or discomfort from system deployment (8 or greater on a 10-point Likert scale)
   - Excessive bleeding
   - Hematoma (slight ecchymosis is a known consequence of needle skin puncture and will not be captured as an AE)
   - Excessive edema from sensor and/or adhesive tape that is significant and non-resolving within 48 hours of sensor pod removal
   - Excessive erythema from sensor and/or adhesive tape that is significant and non-resolving within 48 hours of sensor pod removal
   - Local infection
   - Sensor or introducer needle fracture during insertion/wear/removal

Degrees of edema, erythema, or infection that may occur at the sensor insertion or adhesive tape site will be assessed by study staff. An AE will be recorded as severe in intensity if skin appearance indicates significant edema or erythema (per definition above) and/or if infection, defined as the presence of pus, at the sensor insertion or adhesive tape site occurs.

4.1.2 Recording of Adverse Events

Throughout the course of the study, all efforts will be made to remain alert to possible adverse events or untoward findings. The first concern will be the safety of the study participant, and appropriate medical intervention will be made.

All reportable adverse events whether volunteered by the parent of the participant, discovered by study personnel during questioning, or detected through physical examination, laboratory test, or other means will be recorded on an adverse event form. Because this is a minimal risk, non-intervention study, there will not be a Medical Monitor.

The study investigator will assess the relationship of any adverse event to be related or unrelated by determining if there is a reasonable possibility that the adverse event may have been caused by the study device or study procedures. Sites will be responsible for AE reporting, as per their local IRB policy.

The intensity of adverse events will be rated on a three-point scale: (1) mild, (2) moderate, or (3) severe. It is emphasized that the term severe is a measure of intensity: thus, a severe adverse event is not necessarily serious. For example, itching for several days may be rated as severe, but may not be clinically serious.

Adverse events will be coded using the MedDRA dictionary.

Adverse events that continue after the study participant's discontinuation or completion of the study will be followed until their medical outcome is determined or until no further change in the condition is expected.

4.1.3 Serious Adverse Events

A serious adverse event is any untoward occurrence that:

- Results in death.
- Is life-threatening; (a non-life-threatening event which, had it been more severe, might have become life-threatening, is not necessarily considered a serious adverse event).
- Requires inpatient hospitalization or prolongation of existing hospitalization.
- Results in persistent or significant disability/incapacity or substantial disruption of the ability to conduct normal life functions (sight threatening).
- Is a congenital anomaly or birth defect.
- Could have resulted in death, be life-threatening, or require hospitalization without medical or surgical intervention to prevent any of these events

4.2 Risks and Discomforts

4.2.1 Skin Reactions

The sensor may produce pain when it is inserted into the skin. There is a low risk for developing a local skin infection at the site of the sensor needle placement. Itchiness, redness, bleeding, and bruising at the insertion site may occur as well as local tape allergies.

4.2.2 Blood Draw for Antibodies

There may be pain involved with the blood draw for testing of diabetes antibodies and HbA1c. There is a small risk of bleeding under the skin that will produce a bruise.

4.2.3 Anxiety

It is possible that participants will be diagnosed with diabetes as a part of this study. It is also possible that participants will have indicators that suggest that they are likely to develop diabetes in the future. This knowledge may cause anxiety or concern for participants and parents. If this happens, the clinic personnel will be available to answer any questions the participants or parents may have.

CHAPTER 5: MISCELLANEOUS CONSIDERATIONS

5.1 Benefits

There will be no direct benefit to the participant from being in this study. By helping to establish a normal range of sensor glucose values in non-diabetic individuals, the participant's enrollment in this study is an important contribution to the development of a better understanding of the ability of CGM devices to restore glucose metabolism to normal in patients with diabetes.

5.2 Participant/Parent Reimbursement

Participant compensation will be specified in the informed consent form.

5.3 Participant Withdrawal

Participation in the study is voluntary, and a participant may withdraw at any time. The investigator may withdraw a participant who is not complying with the protocol.

5.4 Confidentiality

For security purposes, participants will be assigned an identifier that will be used instead of their name. Protected health information gathered for this study will be shared with the coordinating center, the Jaeb Center for Health Research in Tampa, FL. During each visit, the study devices will be downloaded to a computer that is secured and password protected and the files will be uploaded to the Coordinating Center via a secure method. All files will include only the participant's identifier; no names or personal information will be included.

Data from the study may be provided to Dexcom, Inc., the company that makes the CGM.

5.5 Discontinuation of the Study

A study participant may be discontinued from the study if the investigator believes that it is not safe for the participant to continue.

The study may be discontinued if recommended by the study investigators for safety or other reasons or if funding for the study is lost.

CHAPTER 6: STATISTICAL CONSIDERATIONS

The approach to sample size and statistical analyses are summarized below. A detailed statistical analysis plan will be written and finalized prior to the completion of the study.

6.1 Sample Size

The sample size is a convenience sample and is not based on statistical principles.

6.2 Data Tabulations

Only participants with: (1) laboratory HbA1c <5.7% and no diabetes antibodies and (2) at least 72 hours of CGM data with at least 24 hours overnight will be included in the analyses.

Results will be tabulated overall and for daytime and overnight as defined by the log.

- Distribution of minimum and maximum glucose level for each participant
- Percentage of participants with one or more sensor values <70/54 mg/dL and percentage of values <70/54 mg/dL
- Percentage of participants with one or more sensor values >140/160/180 mg/dL and percentage of values>140/160/180 mg/dL
- Percentage of values 70-120 and 70-140 mg/dL
- Median and quartiles of glucose concentrations overall
- Standard deviation and/or coefficient of variation

REFERENCES

1. Juvenile Diabetes Research Foundation Continuous Glucose Monitoring Study G, Fox LA, Beck RW, Xing D: Variation of interstitial glucose measurements assessed by continuous glucose monitors in healthy, nondiabetic individuals. Diabetes Care 2010;33:1297-1299

2. Shah VN, DuBose SN, Li Z, Beck RW, Peters AL, Weinstock RS, Kruger D, Tansey M, Sparling D, Woerner S, Vendrame F, Bergenstal R, Tamborlane WV, Watson SE, Sherr J: Continuous Glucose Monitoring Profiles in Healthy Nondiabetic Participants: A Multicenter Prospective Study. J Clin Endocrinol Metab 2019;104:4356-4364